Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $x = e^{\sin^{-1} y}$ then $\left.\frac{dy}{dx}\right|_{x=1}$

(A) -1

(B) 2

(C) $1/2$

(D) 1

Q.2

If $y = \left(x^2 + 1\right)^{\sin x}$, then $y'(0)$ is equal to

(A) 0

(B) 1

(C) 2

(D) -1

Q.3

If $y = \sin\left(n \sin^{-1} x\right)$ satisfies $\left(1 - x^2\right) y'' - xy' + Ay = 0$ then A is equal to

(A) 1

(B) 2

(C) n^2

(D) n

Q.4

If $y = \operatorname{cosec}\left(\cot^{-1} x\right)$, then $\frac{dy}{dx}$ at $x = 1$ is equal to

(A) $\frac{1}{\sqrt{2}}$

(B) 1

(C) $-\sqrt{2}$

(D) $-\frac{1}{\sqrt{2}}$

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Q.5

If $\cos^{-1}\left(\frac{x^2-y^2}{x^2+y^2}\right) = \log a$ then $\frac{dy}{dx}$ is equal to

(A) y/x

(B) x/y

(C) x^2/y^2

(D) y^2/x^2

Q.6

If $y = \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$, then $y'(0)$ is

(A) $1/2$

(B) 0

(C) 1

(D) doesn't exist

Q.7

The derivative of the function $y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ doesn't exist for

(A) all values of x for which $|x| < 1$

(B) $x = -1, 1$

(C) all values of x for which $|x| > 1$

(D) none of these

Q.8

Let $f(x) = \alpha(x)\beta(x)\gamma(x)$ for all real x, where $\alpha(x), \beta(x)$ and $\gamma(x)$ are differentiable functions of x. If $f'(2) = 18f(2), \alpha'(2) = 3\alpha(2), \beta'(2) = -4\beta(2)$ and $\gamma'(2) = k\gamma(2)$, then the value of k is

(A) 14

(B) 16

(C) 19

(D) None of these

Q.9

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A triangle has two of its vertices at $P(a, 0), Q(0, b)$ and the third vertex $R(x, y)$ is moving along the straight line $y = x$. If A be the area of the triangle,
then $\frac{dA}{dx}$ is equal to

(A) $\frac{a-b}{2}$

(B) $\frac{a-b}{4}$

(C) $-\left(\frac{a+b}{2}\right)$

(D) $\frac{a+b}{4}$

Q.10

Let $f(x) = \sqrt{x-1} + \sqrt{x+24-10\sqrt{x-1}}$;
$1 < x < 26$ be real valued function. Then $f'(x)$ for $1 < x < 26$ is

(A) 0

(B) $\frac{1}{\sqrt{x-1}}$

(C) $2\sqrt{x-1} - 5$

(D) None of these

Q.11

If $f(x) = \sqrt{x^2 - 10x + 25}$, then the derivative of $f(x)$ on the interval $[0, 7]$ is

(A) 1

(B) -1

(C) 0

(D) None of these

Q.12

If $f\left(x\right) = \log_{\cot x}\tan x \,.\, \log_{\tan x}\cot x^{-1} + \tan^{-1}\frac{x}{\sqrt{4 - x^2}}, \quad x \in (-2, \ 2)$, then the value of $f'\left(0\right) =$ _____

(A) 0.3

(B) 0.4

(C) 0.5

(D) 0.6

Q.13

If $y = \left(1 + \frac{1}{x}\right)\left(1 + \frac{2}{x}\right)\left(1 + \frac{3}{x}\right) \ldots\ldots \left(1 + \frac{n}{x}\right)$ and $x \neq 0$,
then $\frac{dy}{dx}$ when $x = -1$ is

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(A) $n!$

(B) $(n-1)!$

(C) $(-1)^n(n-1)!$

(D) $(-1)^n n!$

Q.14

f $x<1$, then $\frac{1-2x}{1-x+x^2}+\frac{2x-4x^3}{1-x^2+x^4}+\frac{4x^3-8x^7}{1-x^4+x^8}+\ldots\infty=$

(A) $\frac{1}{1+x+x^2}$

(B) $\frac{1+2x}{1+x+x^2}$

(C) $\frac{1-x+x^2}{1+x+x^2}$

(D) 1

Q.15

The first derivative of the function $\left[\cos^{-1}\left(\sin\sqrt{\frac{1+x}{2}}\right)+x^x\right]$ with respect to x at $x=1$ is

(A) $3/4$

(B) 0

(C) $1/2$

(D) $-1/2$

Answers & Solutions

Q.1 Answer:

1

Solution:

$x=e^{\sin^{-1}y}\Rightarrow\log x=\sin^{-1}y$

$\Rightarrow y=\sin(\log x)$

$\frac{dy}{dx}=\cos(\log x)\frac{1}{x}$

so $\left.\frac{dy}{dx}\right|_{x=1}=\cos(\log 1)=\cos 0=1$

Q.2 Answer:

0

Solution:

Using logarithmic differentiation, we have $\log y=\sin x\left(\log\left(x^2+1\right)\right)$

$\Rightarrow \quad \frac{1}{y}\frac{dy}{dx}=\cos x\left(\log\left(x^2+1\right)\right)+\frac{(\sin x)(2x)}{x^2+1}$

$\Rightarrow \quad \frac{dy}{dx}=\left(x^2+1\right)^{\sin x}\left[\frac{2x\sin x}{x^2+1}+\cos x\left(\log\left(x^2+1\right)\right)\right]$

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So,

$$\left.\frac{dy}{dx}\right|_{x=0} = 1[0 + |\log|]$$
$$= 0$$

Q.3 Answer:

n^2

Solution:

$y' = \cos\left(n\sin^{-1}x\right)\frac{n}{\sqrt{1-x^2}}$

Squaring both the sides, we have

$$\left(1-x^2\right)y'^2 = n^2\cos^2\left(n\sin^{-1}x\right)$$
$$= n^2\left[1-\sin^2\left(n\sin^{-1}x\right)\right]$$
$$= n^2\left[1-y^2\right]$$

Differentiating again $-2xy'^2 + \left(1-x^2\right)2y'y'' = -n^2 2yy'$

$\Rightarrow \quad \left(1-x^2\right)y'' - xy' = -n^2 y$

Hence $A = n^2$

Q.4 Answer:

$\frac{1}{\sqrt{2}}$

Solution:

$\frac{dy}{dx} = -\operatorname{cosec}\left(\cot^{-1}x\right)\cot\left(\cot^{-1}x\right)\frac{d}{dx}\left(\cot^{-1}x\right)$

$= \operatorname{cosec}\left(\cot^{-1}x\right)\times\frac{1}{1+x^2}$

$\left.\frac{dy}{dx}\right|_{x=1} = \operatorname{cosec}\frac{\pi}{4}\times\frac{1}{2} = \sqrt{2}\times\frac{1}{2} = \frac{1}{\sqrt{2}}.$

Q.5 Answer:

y/x

Solution:

$\cos^{-1}\left(\frac{x^2-y^2}{x^2+y^2}\right) = \log a \Rightarrow \frac{x^2-y^2}{x^2+y^2} = \cos$

$\log a = A$ **(say)**

Putting $u = y/x$ **and applying componendo and dividendo, we have**

$(y/x)^2 = u^2 = (1-A)/(1+A)$

$\Rightarrow y/x = \sqrt{(1-A)/(1+A)}$

$\Rightarrow \quad x\,\mathrm{d}y/\mathrm{d}x - y = 0$

$\Rightarrow \mathrm{d}y/\mathrm{d}x = y/x$

Q.6 Answer:

$1/2$

Solution:

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Putting $x = \tan\theta$, we get $\frac{\sqrt{1+x^2}-1}{x} = \frac{\sec\theta-1}{\tan\theta} = \frac{1-\cos\theta}{\sin\theta} = \tan\left(\frac{\theta}{2}\right)$
Hence $y = \frac{1}{2}\tan^{-1}x$ so $y'(x) = \frac{1}{2(1+x^2)}$ and thus
$y'(0) = 1/2.$

Q.7 Answer:

$x = -1, 1$

Solution:

The derivative of $y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ doesn't exist for $\left|\frac{2x}{1+x^2}\right| = 1$. So $2|x| = x^2 + 1$. For $x \geq 0$,
$(x-1)^2 = 0 \Rightarrow x = 1$, if $x < 0$, then $(x+1)^2 = 0$
so $x = -1$.

Q.8 Answer:

19

Solution:

Wehave, $f(x) = \alpha(x)\beta(x)\gamma(x)$, for all real x.
$\Rightarrow f'(x) = \alpha'(x)\beta(x)\gamma(x) + \alpha(x)\beta'(x)\gamma(x)$
$+\alpha(x)\beta(x)\gamma'(x)$
$\Rightarrow f'(2) = \alpha'(2)\beta(2)\gamma(2) + \alpha(2)\beta'(2)\gamma(2)$
$+\alpha(2)\beta(2)\gamma'(2)$
$\Rightarrow 18f(2) = 3\alpha(2)\beta(2)\gamma(2) - 4\alpha(2)\beta(2)\gamma(2)$
$+k\alpha(2)\beta(2)\gamma(2)$
$[\because f'(2) = 18f(2), \alpha'(2) = 3\alpha(2), \beta'(2)$
$= -4\beta(2)$ and $\gamma'(2) = k\gamma(2)]$
$\Rightarrow 18f(2) = (-1+k)\alpha(2)\beta(2)\gamma(2) = (k-1)f(2)$
$[\because f(2) = \alpha(2)\beta(2)\gamma(2)]$
$\Rightarrow k - 1 = 18 \Rightarrow k = 19$

Q.9 Answer:

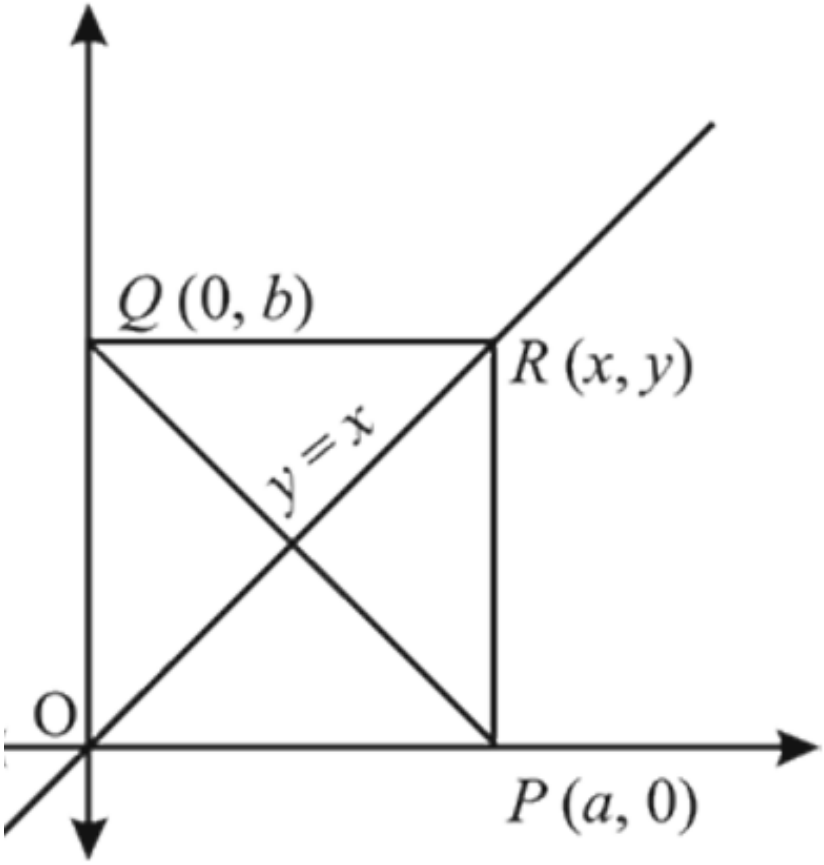
$-\left(\frac{a+b}{2}\right)$

Solution:

Area of $\Delta PQR = A$
$= \frac{\pm 1}{2}[x(b-0) + 0(0-y) + a(y-b)]$
$= \frac{\pm 1}{2}(bx + ax - ab)$

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$\therefore \frac{dA}{dx} = \frac{+1}{2}(a+b)$



Q.10 Answer:

0

Solution:

$f(x) = \sqrt{x-1} + \sqrt{25 + (x-1) - 10\sqrt{x-1}}$
$= \sqrt{x-1} + \sqrt{(5-\sqrt{x-1})^2}$
$= \sqrt{x-1} + |5 - \sqrt{x-1}| = 5$
$[\because \sqrt{x-1} < 5$ **for** $1 < x < 26]$
$\therefore f'(x) = 0$

Q.11 Answer:

None of these

Solution:

We have,

$$f(x) = \sqrt{x^2 - 10x + 25} = \sqrt{(x-5)^2} = |x-5|$$
$$= \begin{cases} x-5, & x \geq 5 \\ 5-x, & x < 5 \end{cases}$$

Clearly, $f(x)$ **is differentiable at all points on the interval** $[0, 7]$ **except at** $x = 5. \therefore$ **The derivative of** $f(x)$ **on the interval** $[0, 7]$ **does tot exist.**

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Q.12 Answer:

0.5

Solution:

Let $a = \log_{\cot\ x}\tan\ x\log_{\tan\ x}\cot\ x^{-1}$

and $b = \tan^{-1}\frac{x}{\sqrt{4-x^2}}$

$\Rightarrow f(x) = a + b$

$a = \log_{\cot x}\tan x\log_{\tan x}\cot x^{-1}$

$= \log_{\cot\ x}\tan\ x\frac{1}{\log_{\tan\ x}\cot\ x}$

$= \frac{\log(\tan\ x)^2}{\log(\cot\ x)} = \frac{\log(\tan\ x)^2}{\log\frac{1}{\tan\ x}}$

$= \frac{\log(\tan\ x)^2}{-\log(\tan\ x)} = 1$

$b = \tan^{-1}\frac{x}{\sqrt{4-x^2}}$

Let$\theta = \sin^{-1}\frac{x}{2}, -2 < x < 2$

$\Rightarrow x = 2\sin\theta$

$\Rightarrow b = \tan^{-1}\frac{2\sin\theta}{\sqrt{4-4\sin^2\theta}}$

$= \tan^{-1}\frac{2\ \sin\ \theta}{2\ \cos\ \theta}$

$= \tan^{-1}(\tan\theta), -\frac{\pi}{2} < \theta < \frac{\pi}{2}$

$= \theta$

$= \sin^{-1}\frac{x}{2}$

$\Rightarrow f(x) = 1 + \sin^{-1}\frac{x}{2}$

$\Rightarrow f'(x) = \frac{1}{\sqrt{1-\frac{x^2}{2}}}\cdot\frac{1}{2}$

$\Rightarrow f'(x) = \frac{1}{\sqrt{4-x^2}}$

$\Rightarrow f'(0) = \frac{1}{\sqrt{4-0}} = \frac{1}{2} = 0.5$

Q.13 Answer:

$(-1)^n(n-1)!$

Solution:

$y = \left(1+\frac{1}{x}\right)\left(1+\frac{2}{x}\right)\left(1+\frac{3}{x}\right)\ldots\left(1+\frac{n}{x}\right)$

$\frac{dy}{dx} = \left(-\frac{1}{x^2}\right)\left(1+\frac{2}{x}\right)\left(1+\frac{3}{x}\right)\cdots\left(1+\frac{n}{x}\right)$

$+\left(1+\frac{1}{x}\right)\left(-\frac{2}{x^2}\right)\left(1+\frac{3}{x}\right)\ldots\left(1+\frac{n}{x}\right)$

$+\ldots+\left(1+\frac{1}{x}\right)\left(1+\frac{2}{x}\right)\left(1+\frac{3}{x}\right)\ldots\left(-\frac{n}{x^2}\right)$

$\because \frac{dy}{dx}\Big|_{x=1} = (-1)(-1)(-2)(-3)\ldots.(1-n)$

$= (-1)^n(1)(2)(3)\ldots.(n-1) = (-1)^n(n-1)!$

Q.14 Answer:

$\frac{1+2x}{1+x+x^2}$

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Solution:

We have,

$\left(1+x+x^2\right)\left(1-x+x^2\right)\left(1-x^2\right)^2-x^2=1+x^2+x^4$

Now, $\left(1+x+x^2\right)\left(1-x+x^2\right)\left(1-x^2+x^4\right)=\left(1+x^2+x^4\right)\left(1-x^2+x^4\right)=1+x^4+x^8$

Continuing in this way, we have

$\left(1+x+x^2\right)\left(1-x+x^2\right)\left(1-x^2+x^4\right)\left(1-x^4+x^8\right) \quad \ldots\left(1-x^{2^{n-1}}+x^{2^n}\right)$

$=\left(1+x^{2^n}+x^{2^{n+1}}\right)$

Now, for $x<1, x^\infty=0$. **Taking limits as** $n \to \infty$ **in** (1)**, we get**

$\left(1+x+x^2\right)\left(1-x+x^2\right)\left(1-x^2+x^4\right)\left(1-x^4+x^8\right) \ldots=1$

Taking logarithm of both sides, we get

$\Rightarrow \ln\left(1+x+x^2\right)+\ln\left(1-x+x^2\right)+\ln\left(1-x^2+x^4\right)+\ln\left(1-x^4+x^8\right)+\ldots=0$

Differentiating both sides w.r.t. x**, we get**

$\Rightarrow \frac{1+2x}{1+x+x^2}+\frac{-1+2x}{1-x+x^2}+\frac{-2x+4x^3}{1-x^2+x^4}+\frac{-4x^3+8x^7}{1-x^4+x^8}+\ldots=0$

Hence, $\frac{1-2x}{1-x+x^2}+\frac{2x-4x^3}{1-x^2+x^4}+\frac{4x^3-8x^7}{1-x^4+x^8}+\ldots=\frac{1+2x}{1+x+x^2}$

Q.15 Answer:

$3/4$

Solution:

$f(x)=\cos^{-1}\left[\cos\left(\frac{\pi}{2}-\sqrt{\frac{1+x}{2}}\right)\right]+x^x f(x)=\frac{\pi}{2}-\sqrt{\frac{1+x}{2}}+x^x$

$\Rightarrow f'(x)=-\frac{1}{\sqrt{2}} \times \frac{1}{2\sqrt{1+x}}+x^x(1+\log x)$

$f'(1)=-\frac{1}{4}+1=\frac{3}{4}$

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